

# JOSEPH M. HILBY

[Joseph.Hilby@gmail.com](mailto:Joseph.Hilby@gmail.com) • 480.209.9613 • Pensacola, FL • [LinkedIn](#) • [GitHub](#) • Veteran • Security Clearance: Secret

Former U.S. Navy Lieutenant and Aviator with 10+ years leading 60+ personnel in safety-critical aviation operations, pilot training, aircraft maintenance, and large-scale logistics. Directly responsible for more than \$1B in aircraft, supporting the evacuation of Afghanistan, Hurricane Maria humanitarian relief, and multiple Arabian Gulf deployments. Currently finishing a second bachelors degree in Computer Science in order to build a strong technical foundation to succeed and lead in emerging aerospace and defense technologies.

## EXPERIENCE

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**Skills:** Logistics • Training • Standardization • Process Improvement • Risk Management • Emergency Response • Disaster Relief

**Reserve Officer, Naval Reserve Center Pensacola** 2021 - 2023

**Assistant Air Officer, USS IWO JIMA (LHD-7)** 2020 - 2021

- Responsible for overall shipboard aviation operations, aircraft maintenance, personnel leadership, and long-term resource planning.
- Led 40-person team that managed the operation and maintenance of four aircraft models and 25 aircraft deployed on USS Iwo Jima.
- Accountable officer for flightdeck operations and safety. Personally supervising all flight deck operations and personnel, including 20 takeoffs/landings per hour, pilots, air traffic controllers, aircraft maintainers, loading/unloading crew, and combat personnel.
- Supported large-scale humanitarian evacuation and emergency operations during the Evacuation of Afghanistan.

**Flight Instructor & Standardization Check Pilot, Training Squadron TEN (VT-10)** 2018 - 2020

- Trained +250 Naval Flight Officer students from the U.S., Italy, Germany, and Saudi Arabia to fly fixed-wing military aircraft; to respond to emergencies, e.g., engine and other critical system failures; and to conduct mission critical combat responsibilities, including navigation, route planning, aircraft formation, communication, and fuel management.
- Awarded 2019 Flight Instructor of the Year for Command Training Wing SIX and Training Squadron TEN.

**Flight Instructor & Maintenance Check Pilot, Helicopter Maritime Strike Squadron SEVEN TWO (HSM-72)** 2011 - 2018

- Flew 100+ helicopter missions during 10-month Arabian Gulf deployment with carrier strike group. Missions included search and rescue; intelligence, surveillance and reconnaissance; and antisubmarine and antisurface warfare.
- One of three officers managing and accountable for all pre-deployment logistics for squadron's deployment to Arabian Gulf, including planning and movement of all materials, aircraft maintenance training and qualification of 60+ personnel, and related documentation.
- Selected by the commanding officer to serve as the squadron Standardization Officer, aka "NATOPS Officer," responsible for annually assessing and approving all squadron pilots, including the commanding officer, as ready and safe to fly.
- Maintenance Check Pilot responsible for test flying all newly repaired squadron aircraft to validate safe-for-flight certification. Aircraft overseen consisted of 13 squadron helicopters with an approximate cumulative value of \$1 billion.
- Planned and executed emergency, humanitarian aid and disaster relief missions in Florida Keys following Hurricane Maria in 2017.

## EDUCATION

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**University of Florida - BS, Computer Science** 2022 - 2026

- Tutor for undergraduate computer science, mathematics & physics at Pensacola State College

**Turing School of Software & Design - Certificate, Backend Software Engineering** 2021 - 2022

**The College of Idaho - BA, Philosophy** 2006 - 2010

- Treasurer, Delta Tau Delta; Idaho Kendo Club

## COMPUTER SCIENCE PROJECTS

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**Skills:** C/C++ • Computer Architecture • Java • Compiler Design • Python • Data Science

[FreshlyGround](#) | **Java, TypeScript | Custom Programming Language & Compiler**

- Designed and implemented lexical evaluation, syntax parsing, intermediate representation construction, semantic analysis, and assembly generation.
- Compiler targeting WebAssembly to enable a web IDE for user interface.

[Hack++](#) | **C, JavaScript | Custom CPU, Compiler, and Operating System**

- Implemented a complete computer system from first principles, spanning logic gates, CPU, memory, and I/O.
- Defined the instruction set architecture and hardware-software interfaces to ensure separation of concerns across system layers.
- Created a frontend web interface for users to interact with the system.